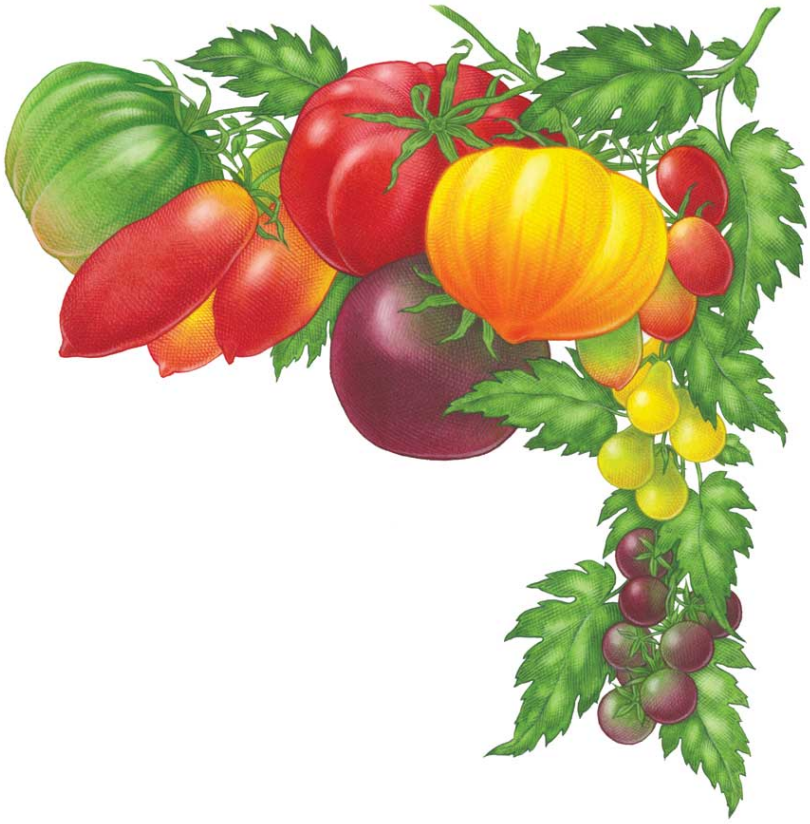
MOTHER EARTH NEWS

All About Growing Tomatoes

Learn how to plant, grow and harvest tomatoes, plus peruse our recommended tomato varieties.
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By Barbara Pleasant



Try growing some of these beautiful tomato varieties in your vegetable garden (from left to right): 'Green Zebra,' 'San Marzano,' 'Brandywine,' 'Cherokee Purple,' 'Mr. Stripey,' 'Juliet,' 'Yellow Pear' and 'Black Cherry.'

Illustration by Keith Ward

(For details on growing many other vegetables and fruits, visit our Crop at a Glance collection page.)

The exquisite flavor and irresistible juiciness of homegrown tomatoes put them at or near the top of most gardeners' planting lists. Fruit size, color and flavor differ with each variety, but all tomatoes grow best under warm conditions. For the best flavor,

provide fertile, organically enriched soil with a pH between 6.0 and 6.5, and plant your tomatoes in a site that gets plenty of sun.

Tomato Types to Try

Many organic gardeners include varieties of the following three types of tomatoes in their gardens each year.

Cherry tomatoes and **salad tomatoes** produce small fruits in a rainbow of colors and an array of shapes, including round, pear-shaped and teardrop-shaped.

Slicing tomatoes are round and juicy, making them ideal for eating fresh. Fruit size and color vary, and some varieties produce surprisingly large fruits.

Paste tomatoes, also called canning tomatoes, have dense flesh and little juice, so they are the best type for cooking, canning and drying.

See our chart for more information on these types, including recommended varieties.

When to Plant Tomatoes

Start seeds indoors under bright fluorescent lights in early spring, about six to eight weeks before your last spring frost. If kept moist and warm (above 70 degrees Fahrenheit), tomato seeds should sprout within a week. Transplant the seedlings to larger containers when they are about 6 weeks old. Harden off seedlings by gradually exposing them to outdoor weather for a few hours each day for at least a week before transplanting them. Transplant seedlings to the garden (or to large containers) after your last frost has passed, during a period of warm weather.

How to Plant Tomatoes

Choose a sunny site with fertile, well-drained soil, and loosen the planting bed to 12 inches deep. Mix in a layer of mature compost. Dig planting holes at least 18 inches apart, and enrich each with a spadeful of additional compost mixed with a balanced organic fertilizer (follow application rates on the fertilizer label). Plant tomatoes deeper than they grew in their containers, so that only the top five or six leaves show at the surface. Additional roots will grow from the buried section of stem.

Tomato Harvesting and Storage

Tomato flavor declines at temperatures below 55 degrees, so never keep them in the refrigerator. If kept in a warm place, fruits picked when they're showing stripes or blushes of ripe color will continue to ripen. Bumper crops can be preserved by canning, drying or freezing. Tomatoes don't need to be blanched before they are dried or frozen.

Tomato sauce, salsa, chutney and ketchup can be processed in a water bath canner or a pressure canner, depending on acidity levels (always follow recipes when canning). You will need a pressure canner to can tomato soup that includes other vegetables.

Saving Tomato Seeds

To save seeds from open-pollinated varieties, allow perfect fruits to ripen until they become soft. Cut them in half and squeeze the gel and seeds into a small jar. Cover with 3 inches of water, and shake well. Allow the mixture to sit at room temperature for 24 hours before pouring out the liquid. Rinse the big seeds at the bottom of the jar in a strainer, and then dry them on a paper plate for about two weeks (write the variety name on the plate). If handled this way and given cool, dry storage conditions, tomato seeds usually stay viable for four to six years.

If you're growing tomatoes for seed-saving, keep in mind that wind and insects can transfer pollen, creating crosses between varieties. For pure seed, save seeds from plants that were grown apart from other tomato varieties.

Preventing Tomato Pests and Diseases

Most tomatoes are susceptible to a fungal disease called early blight (*Alternaria solani*), which develops in early summer and causes leaves near the ground to develop dry, brown patches surrounded by concentric, black rings. Good genetic resistance is not available, although 'Plum Dandy,' a Roma type, offers noticeable tolerance. (Check our Seed and Plant Finder to find 'Plum Dandy' sources.) The best intervention is to prune off affected leaves as soon as you see them. Removing all leaves within 18 inches of the ground can reduce or delay outbreaks. Most tomato plants produce well despite losing leaves to early blight.

Hard black or brown patches on the blossom ends of ripening tomatoes indicate a physiological disorder called blossom-end rot, which is most common in large-fruited varieties. Prevent this problem by growing tomatoes in fertile soil generously enriched with compost, and mulch heavily to keep soil moisture levels as constant as possible.

Tomato hornworms (large, green caterpillars with white stripes) are the larvae of a large moth. Handpick them starting in early summer (follow the trail of pebbly caterpillar droppings to find them). In extremely bad years, control hornworms using Bt (*Bacillus thuringiensis*) or spinosad, two widely available biological pesticides.

The devastating fungal disease called late blight (*Phytophthora infestans*) may strike following a prolonged period of cool rain. Affected leaves develop light brown, water-soaked patches, and entire plants can wilt within a few days. Provide excellent light penetration and air circulation to keep plants dry, reducing the risk of late blight. Reliable genetic resistance is not available, though a few varieties, such as 'Juliet,' have shown some tolerance. In climates where late blight is a problem every year, try growing tomatoes in high tunnels — the best way to keep the plants dry.

Tomato Growing Tips and Ideas

Stake or cage tomatoes to raise them above damp conditions close to the ground. Cages support plants from all sides, so they are preferred for large, vigorous varieties. All tomato cages become top-heavy after the plants set fruit, so plan to anchor them to sturdy stakes to prevent toppling. For ideas and plans for do-it-yourself supports, see Build Homemade Tomato Cages.

Include early and midseason varieties in your garden to ensure a long harvest season. In long-season regions, root the stem tips from healthy plants in early summer, and they'll quickly grow new plants for a fall crop.

Prevent cracked fruits and blossom-end rot by mulching tomatoes heavily in early summer, after the soil has warmed. Place a soaker hose beneath the mulch if summer droughts are common in your area.

For recommended seed-sowing and transplanting dates for your local climate, try our new Vegetable Garden Planner.

In the Kitchen

Tomato sandwiches dripping with juice capture tomato cuisine at its purest, but summer wouldn't be complete without salsa, tomato basil soup, or hot pasta tossed with chopped tomatoes, basil, garlic and a good hard cheese.

Tomatoes that ripen to black, green, orange or yellow add visual appeal to dishes and often have unusual textures and flavors, ranging from the firm and citrusy 'Green Zebra' to the soft and smoky 'Black Krim.' Use green tomatoes gathered at season's end to make tomato chutney or other condiments to enjoy year-round.

All tomatoes are good sources of vitamins A and C, and red tomatoes provide lycopene, a powerful antioxidant that may help prevent some types of cancer. Green tomatoes also have high nutritional value.

Contributing editor Barbara Pleasant *gardens in southwest Virginia, where she grows vegetables, herbs, fruits, flowers and a few lucky chickens. Contact Barbara by visiting* her website *or finding her on* Google+.